
بسم الله الرحمن الرحیم

Image Quality Assessment & Statistical Evaluation

Image Quality Assessment-Errors

- **Many remote sensing datasets contain high-quality, accurate data**
- **Sometimes error (or noise) is introduced into the remote sensor data by:**
  - **The environment (e.g., atmospheric scattering)**
  - **Random or systematic malfunction of the remote sensing system**
  - **Improper airborne or ground processing of the remote sensor data prior to actual data analysis**

Image Quality Assessment & Statistical Evaluation

- **The image analyst responsible for analyzing the digital remote sensor data should first assess its quality and statistical characteristics.**
- **This is normally accomplished by:**
  - **Looking at the frequency of occurrence of individual brightness values in the image displayed in a histogram**
  - **Viewing on a computer monitor individual pixel brightness values at specific locations or within a geographic area**

Image Quality Assessment & Statistical Evaluation

- **Computing univariate descriptive statistics to determine if there are unusual anomalies in the image data**
- **Computing multivariate statistics to determine the amount of between-band correlation (e.g., to identify redundancy)**

Remote Sensing Sampling Theory

- ***Population*** **: an infinite or finite set of elements**
  - A population could be all possible images that might be acquired of the Earth in year 2025
- ***Sample:*** **a subset of the elements taken from a population used to make inferences about certain characteristics of the population**
  - For example, we might decide to analyze a June 1, 2025, Landsat image of AOI

Remote Sensing Sampling Theory

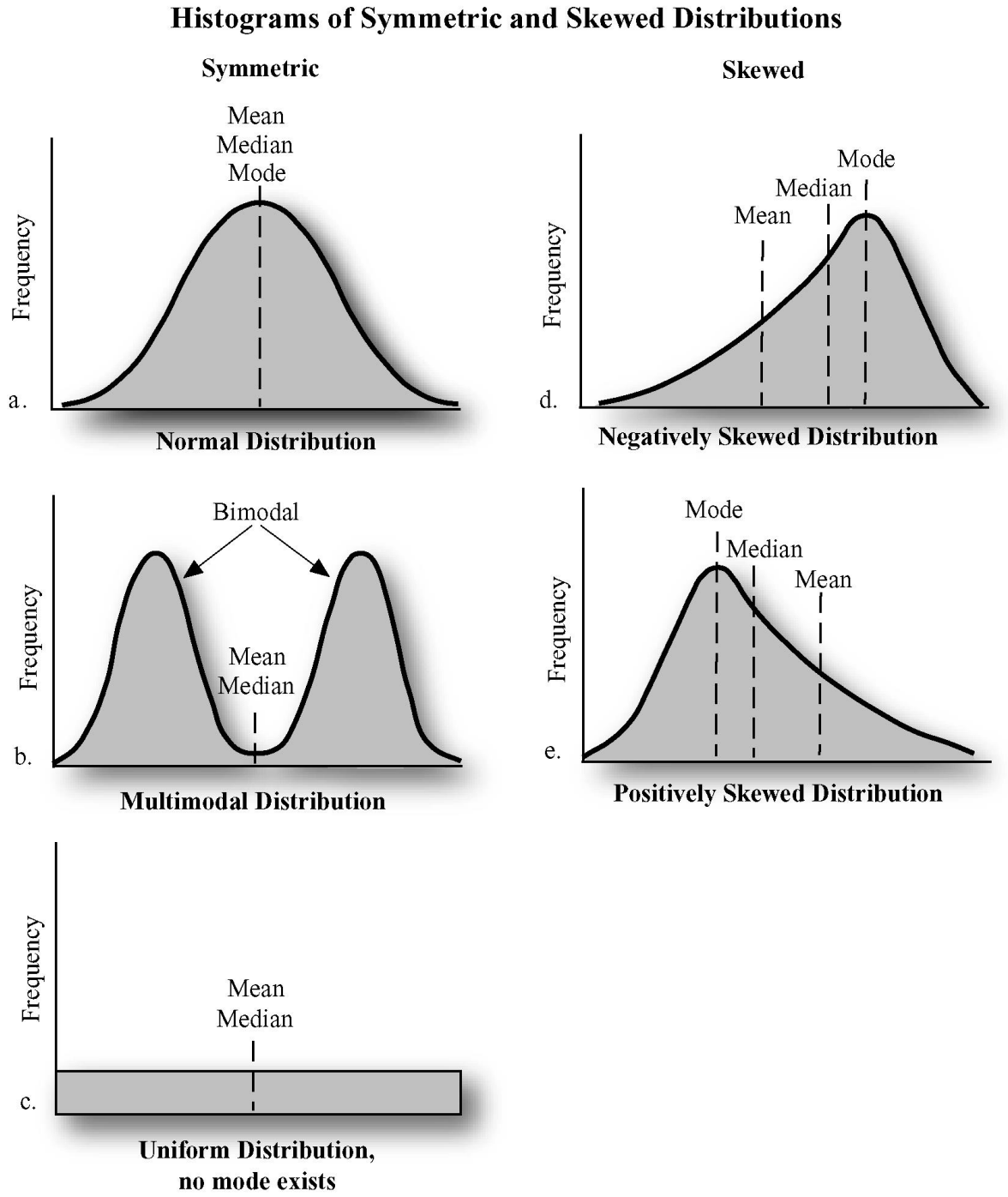
- ***Biased* sample:** If observations with certain characteristics are systematically excluded from the sample either deliberately or inadvertently
  - **such as selecting images obtained only in the summer season**
- ***Sampling error:***
  - The difference between the true value of a population characteristic and the value of that characteristic inferred from a sample

Remote Sensing Sampling Theory

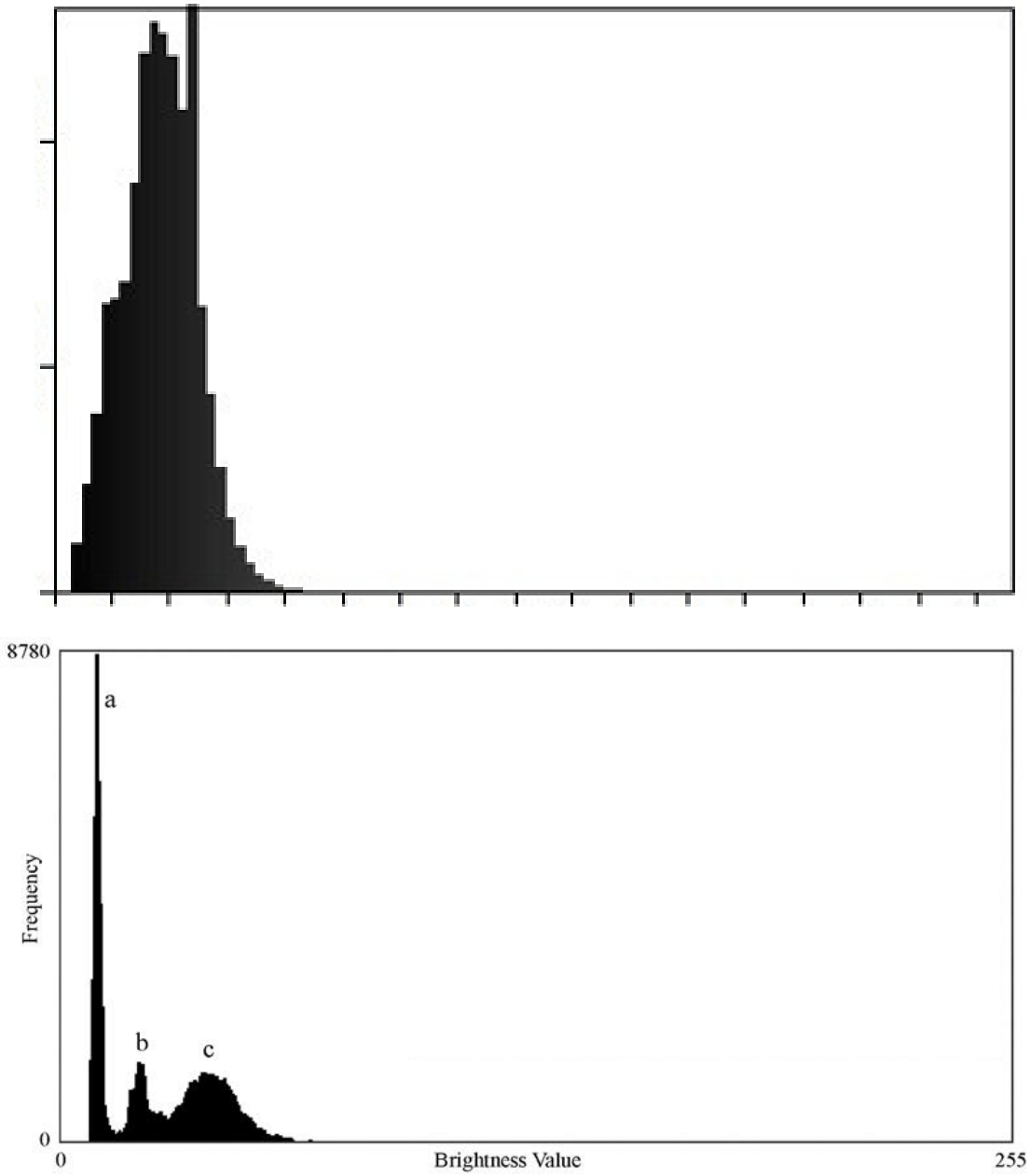
- **Large samples drawn randomly from natural populations usually produce a *symmetrical frequency distribution***
- **Most values are clustered around some central value, and the frequency of occurrence declines away from this central point.**
- **A graph of the distribution appears bell-shaped and is called a *normal distribution***
- **Many statistical tests used in the analysis of remotely sensed data assume that the brightness values recorded in a scene are *normally distributed***

Histogram

- Image histogram provides initial information about the most important parameter of an image
- Histogram is one of the simplest but important and useful component in digital image processing
- It is a distribution function which provides the number of pixels having a particular Digital Number
- It can describe an image just in statistical terms without explaining its spatial patterns



Common Symmetric and Skewed Distributions in Remotely Sensed Data

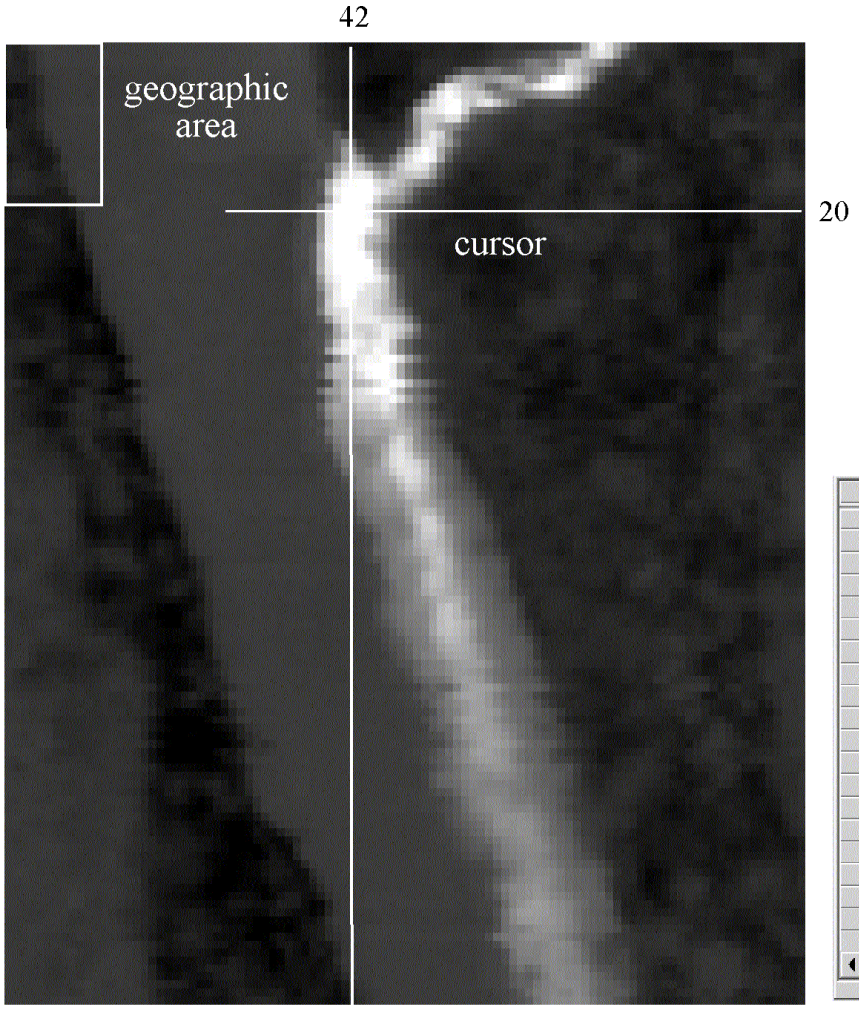


a. Histogram of the original Landsat Thematic Mapper Band 4 imagery.

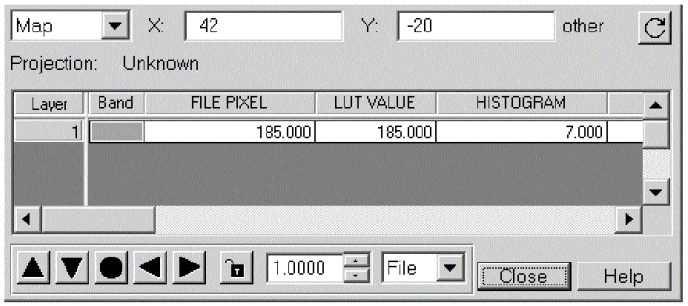
Viewing Individual Pixels

- *Viewing individual pixel brightness values* in a remotely sensed image is one of the useful methods for assessing the quality and information content of the data.
- Use a mouse-controlled cursor (cross-hair) to identify a geographic location in the image (at a particular row and column or geographic x,y coordinate) and display its brightness value in n bands
  - Display the individual brightness values of an individual band in a matrix (raster) format

Cursor and Raster Display of Brightness Values



a. Nightime thermal infrared image of the Savannah River.



b. Individual brightness value extracted using the cursor.

Row	0	1	2	3	4	5	6	7	8	9	10	11
0	63	61	67	73	76	77	75	76	76	76	76	76
1	67	62	62	68	73	76	75	76	76	76	76	76
2	65	62	61	67	73	76	75	76	76	76	76	76
3	63	63	62	67	72	75	75	75	75	75	75	76
4	61	62	61	61	67	73	74	75	75	75	75	74
5	56	53	55	60	67	72	74	75	75	75	75	74
6	53	54	55	59	67	72	74	74	75	75	74	75
7	58	60	58	59	65	72	74	75	75	75	75	75
8	56	59	58	61	68	72	74	75	76	76	76	76
9	55	60	60	62	68	74	74	74	75	75	75	75
10	59	61	60	61	66	73	74	74	75	75	75	75
11	64	62	60	58	61	69	75	75	75	75	75	75
12	63	62	58	56	58	65	72	74	74	75	75	75
13	61	62	60	58	56	58	66	74	74	74	75	75
14	57	60	61	60	57	56	61	71	74	74	74	74
15	61	64	63	59	56	58	65	73	74	74	75	75
16	63	66	67	64	61	61	63	68	73	76	75	74
17	65	68	67	64	62	63	63	67	74	75	75	75
18	64	66	65	64	62	59	58	61	71	75	75	75
19	60	60	59	58	58	59	59	58	62	72	74	73

c. Brightness values in a geographic area for an individual band.

Univariate Descriptive Image Statistics

Band Statistics

- **The distribution of data values in a single band could be represented graphically using a histogram and mathematically by the variance statistics, which summarizes the differences between all the pixel values and the mean value of each band**

Univariate Descriptive Image Statistics

- **Measures of Central Tendency in Remote Sensor Data**
  - **Mean**
  - **Median**
  - **Mode**
  - **Range**
  - **Variance**
  - **Standard Deviation**

Univariate Descriptive Image Statistics

- ***Mean*** **is the arithmetic average (the mean brightness of an image)**
- **Sum of all brightness value observations divided by the number of observations**

$$\mu_k = \frac{\sum_{i=1}^{n} BV_{ik}}{n}$$

Univariate Descriptive Image Statistics

- **The sample mean is an unbiased estimate of the population mean if it tends to be closer to the population mean**

- **The sample mean is a poor measure of central tendency when the set of observations is skewed or contains an outlier**

Univariate Descriptive Image Statistics

- ***Mode:***
  - ***The value that occurs most frequently in a distribution*** **and is usually the highest point on the curve (histogram)**
  - **There can be more than one mode in a remote sensing dataset**
  - **More than one modes data have non-symmetrical (skewed) distributions**

Univariate Descriptive Image Statistics

- **Median**
  - ***The value midway in the frequency distribution***
  - **One-half of the area below the distribution curve is to the right of the median, and one-half is to the left**

Remote Sensing Univariate Statistics - **Range**

- ***Range*** **of a band of imagery is the difference between the *maximum* and *minimum* values**
- **When the minimum or maximum values are extreme or unusual observations (*data blunders*), the range could be a misleading measure of dispersion**
- **Such extreme values may occur due to the remote sensor's delicate electronics and other malfunctions**

Variance

- **The variance is a means to express the variation of pixel values within a single raster map (a band)**

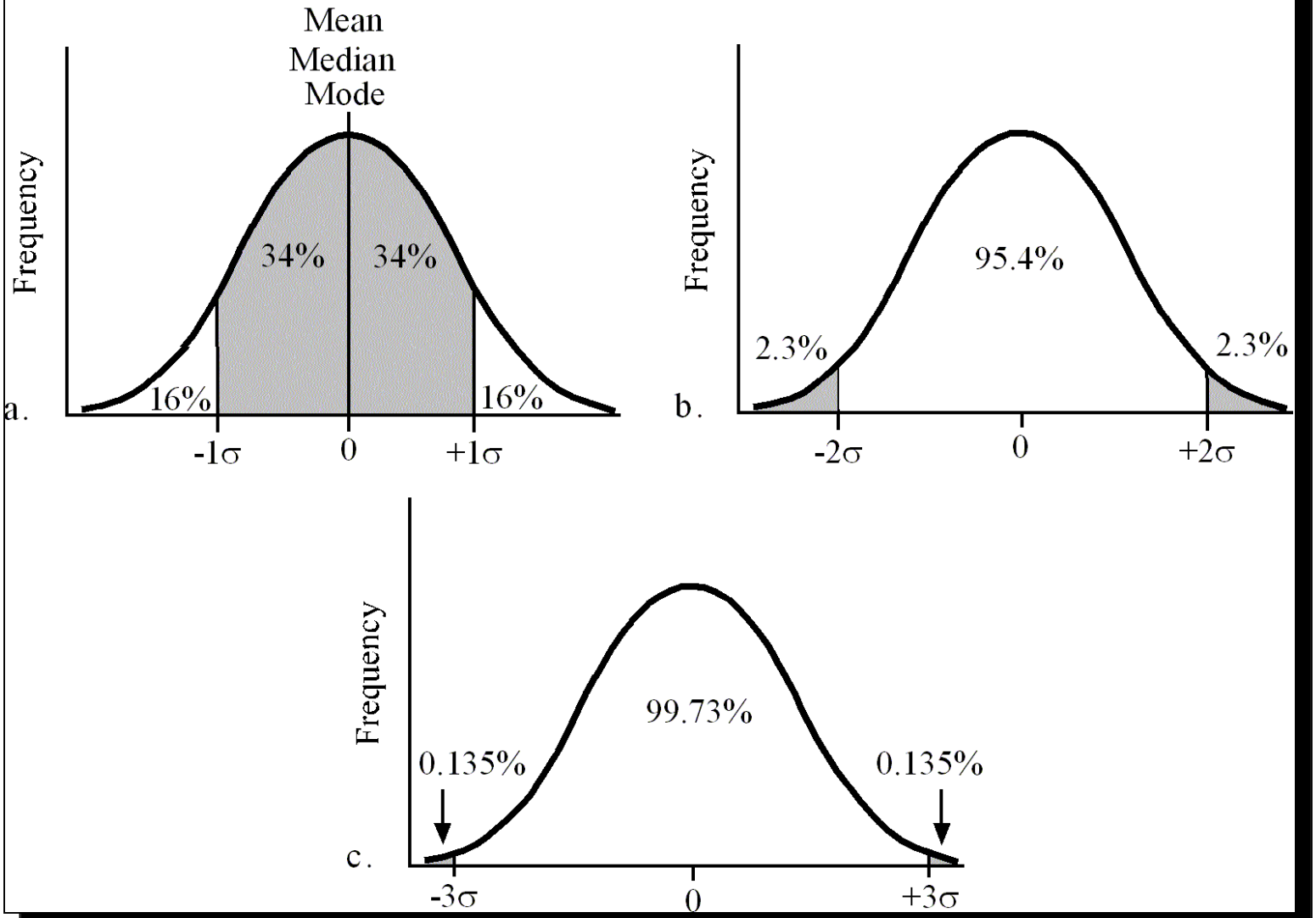
$$\mathrm{var}_k = \frac{\sum_{i=1}^{n} \left(BV_{ik} - \mu_k \right)^2}{n}$$

Standard deviation

- **Standard deviation**
  - **The tendency of dispersion of a distribution from the center**
  - **It is the positive square root of the variance**

$$\sigma_k = \sqrt{var_k}$$

Areas Under the Normal Curve for Various Standard Deviations from the Mean



Hypothetical Dataset of Brightness Values

Pixel	Band 1 (green)	Band 2 (red)	Band 3 (near-infrared)	Band 4 (near-infrared)
(1,1)	130	57	180	205
(1,2)	165	35	215	255
(1,3)	100	25	135	195
(1,4)	135	50	200	220
(1,5)	145	65	205	235

Univariate Statistics for a Hypothetical Sample Dataset

	Band 1 (green)	Band 2 (red)	Band 3 (near-infrared)	Band 4 (near-infrared)
Minimum (min_k)	**100**	**25**	**135**	**195**
Maximum (max_k)	**165**	**65**	**215**	**255**
Range (BV_r)	**65**	**40**	**80**	**60**
Mean (m_k)	**135**	**46.40**	**187**	**222**
Variance (var_k)	**562.50**	**264.80**	**1007**	**570**
Standard deviation (σ_k)	**23.71**	**16.27**	**31.4**	**23.87**

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